04_data_visualization

February 27, 2026

1 Data Visualization for Actuaries

1.1 A 90-Minute Practical Workshop

Welcome! In this session, we will explore essential data manipulation and visualization techniques using Python. You will learn to work with NumPy arrays, manipulate data with Pandas, and create compelling visualizations using Pandas and Seaborn.

1.1.1 Session Overview

- **Part 1**: NumPy Basics — Arrays, indexing, and vectorized operations
- **Part 2**: Pandas Basics — DataFrames, data exploration, and aggregation
- **Part 3**: Plotting with Pandas — Creating informative visualizations
- **Part 4**: Advanced Visualization — Seaborn distributions and pairplots

Duration: 90 minutes
Audience: Actuaries and quantitative professionals
Prerequisites: Basic Python knowledge

[1]:
```
# Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from mytools import generate_sample_insurance_data

# Set visualization defaults
sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['font.size'] = 11
```

1.2 Part 1: NumPy Basics

NumPy is the foundation of numerical computing in Python. It provides efficient arrays and vectorized operations that are essential for data analysis and scientific computing.

1.2.1 1.1 Creating Arrays from Nested Lists

You can create NumPy arrays from Python lists. NumPy arrays can be multi-dimensional:

Dimension	Example	Description
1D	`np.array([1, 2, 3])`	A single row or column
2D	`np.array([[1, 2], [3, 4]])`	A matrix with rows and columns
3D+	Higher-dimensional structures	For complex data

Key advantages of NumPy arrays: - Efficient memory usage - Fast vectorized operations - Broadcasting across different array shapes - Seamless integration with other libraries

1.2.2 1.2 Array Indexing and Slicing

Once you have an array, you can access elements in multiple ways:

```
arr = np.array([10, 20, 30, 40, 50])
arr[0]        # First element: 10
arr[-1]       # Last element: 50
arr[1:4]      # Elements from index 1 to 3: [20, 30, 40]
arr[::2]      # Every second element: [10, 30, 50]
```

For 2D arrays, you use `arr[row, column]` or `arr[row][column]`.

1.2.3 1.3 Universal Functions (ufuncs)

Universal functions perform element-wise operations on arrays. Common ufuncs include:

- `np.sqrt()` — Square root
- `np.exp()` — Exponential function
- `np.log()` — Natural logarithm
- `np.sin()`, `np.cos()` — Trigonometric functions
- `np.abs()` — Absolute value

These operations are **vectorized**, meaning they apply to all elements simultaneously without explicit loops.

1.2.4 Example: Insurance Premium Calculation

Suppose you have an array of base premiums and want to apply a 10% loading. With NumPy:

$$\text{Loaded Premium} = \text{Base Premium} \times 1.10$$

[2]:
```
# Example: Creating and manipulating NumPy arrays
base_premiums = np.array([500, 750, 1000, 1200, 950])

# Apply a 10% loading
loaded_premiums = base_premiums * 1.10

print("Base Premiums:", base_premiums)
print("Loaded Premiums:", loaded_premiums)

# Create a 2D array (matrix) of premium data
```

```python
# Rows: policyholders, Columns: base, loading, final
premium_data = np.array([
    [500, 50, 550],
    [750, 75, 825],
    [1000, 100, 1100],
    [1200, 120, 1320],
    [950, 95, 1045]
])

print("\nPremium Data (2D Array):")
print(premium_data)

# Access individual elements
print("\nFirst policyholder's final premium:", premium_data[0, 2])

# Access a column (all final premiums)
print("All final premiums:", premium_data[:, 2])

# Access a row (first policyholder's data)
print("First policyholder's data:", premium_data[0, :])
```

```
Base Premiums: [ 500  750 1000 1200  950]
Loaded Premiums: [ 550.  825. 1100. 1320. 1045.]

Premium Data (2D Array):
[[ 500   50  550]
 [ 750   75  825]
 [1000  100 1100]
 [1200  120 1320]
 [ 950   95 1045]]

First policyholder's final premium: 550
All final premiums: [ 550  825 1100 1320 1045]
First policyholder's data: [500  50 550]
```

1.2.5 1.4 Exercise: NumPy Fundamentals

Complete the following tasks:

1. Create a 1D array of claim amounts: `[100, 250, 500, 1500, 350]`
2. Apply a logarithmic transformation using `np.log()` (useful for skewed claim data)
3. Create a 2D array of risk factors and premiums
4. Extract and manipulate specific elements using indexing

Instructions: - Fill in the code cells below to complete each task - Experiment with different indexing and slicing techniques - Verify your results by printing the output

[3]:
```python
# Task 1: Create a 1D array of claim amounts and apply log transformation
claim_amounts = np.array([100, 250, 500, 1500, 350])
```

```python
log_claims = np.log(claim_amounts)

print("Original claim amounts:", claim_amounts)
print("Log-transformed claims:", log_claims)

# Task 2: Create a 2D array with policyholder risk factors and corresponding␣
↪premiums
# Columns: [risk_factor, base_premium, adjusted_premium]
risk_premium_data = np.array([
    [1.0, 500, 500],
    [1.15, 500, 575],
    [1.35, 500, 675],
    [2.0, 500, 1000],
    [0.8, 500, 400]
])

print("\nRisk Premium Data:")
print(risk_premium_data)

# Task 3: Extract specific elements
# Get the adjusted premium for the third policyholder
third_premium = risk_premium_data[2, 2]
print(f"\nThird policyholder's adjusted premium: {third_premium}")

# Task 4: Extract all adjusted premiums (last column)
all_adjusted = risk_premium_data[:, 2]
print(f"All adjusted premiums: {all_adjusted}")

# Bonus: Apply a universal function - calculate square root of risk factors
risk_factors = risk_premium_data[:, 0]
sqrt_risk = np.sqrt(risk_factors)
print(f"\nSquare root of risk factors: {sqrt_risk}")
```

```
Original claim amounts: [ 100  250  500 1500  350]
Log-transformed claims: [4.60517019 5.52146092 6.2146081  7.31322039 5.85793315]

Risk Premium Data:
[[1.00e+00 5.00e+02 5.00e+02]
 [1.15e+00 5.00e+02 5.75e+02]
 [1.35e+00 5.00e+02 6.75e+02]
 [2.00e+00 5.00e+02 1.00e+03]
 [8.00e-01 5.00e+02 4.00e+02]]

Third policyholder's adjusted premium: 675.0
All adjusted premiums: [ 500.  575.  675. 1000.  400.]

Square root of risk factors: [1.         1.07238053 1.161895   1.41421356
0.89442719]
```

1.3 Part 2: Pandas Basics

Pandas is built on top of NumPy and provides the `Series` and `DataFrame` objects. A DataFrame is like a spreadsheet with labeled columns and rows, making data manipulation intuitive.

1.3.1 2.1 Creating Series from Dictionaries

A Series is a one-dimensional labeled array. You can create it from a dictionary:

```
age_series = pd.Series({'Alice': 28, 'Bob': 35, 'Carol': 42})
```

The dictionary keys become the index, and the values become the data.

1.3.2 2.2 Creating DataFrames

You can combine multiple Series to form a DataFrame:

```
df = pd.DataFrame({
    'age': [28, 35, 42],
    'premium': [500, 750, 1000],
    'region': ['North', 'South', 'East']
})
```

Or from a dictionary of Series, ensuring all have the same index.

1.3.3 2.3 Accessing Data

Pandas provides multiple ways to access data:

Method	Purpose	Example
`.loc[]`	Label-based access	`df.loc[0, 'age']`
`.iloc[]`	Position-based access	`df.iloc[0, 1]`
`df[column]`	Access entire column	`df['age']`
`df.at[]`	Fast scalar access	`df.at[0, 'age']`

1.3.4 2.4 Data Exploration Methods

Pandas provides convenient methods for understanding your data:

```
df.head()            # First 5 rows
df.tail()            # Last 5 rows
df.info()            # Data types and non-null counts
df.describe()        # Statistical summary
df.columns           # Column names
df.dtypes            # Data types
df.isna().sum()      # Count missing values per column
df['column'].unique()       # Unique values
df['column'].value_counts() # Value frequencies
```

1.3.5 2.5 Grouping and Aggregation

A powerful feature for actuarial analysis is grouping:

`df.groupby('region')['premium'].mean()`

This groups by region and calculates the mean premium per region.

[4]:
```
# Load sample insurance data
df = generate_sample_insurance_data(n=300, seed=42)

# Display first few rows
print("First 5 rows of the dataset:")
print(df.head())

# Display data info
print("\n" + "="*50)
print("Dataset Information:")
print("="*50)
df.info()

# Display statistical summary
print("\nStatistical Summary:")
print(df.describe())
```

```
First 5 rows of the dataset:
   policyholder_id  age  gender   region vehicle_type  annual_premium  \
0                1   56  Female     East    Mid-range         1755.21
1                2   69    Male     West    Mid-range         1344.78
2                3   46  Female  Central          SUV          590.09
3                4   32    Male     West      Compact         1417.90
4                5   60  Female     East       Luxury         1482.04

   claims_count  claim_amount
0             1        811.11
1             0          0.00
2             2       2594.37
3             3       1761.43
4             1        641.49

==================================================
Dataset Information:
==================================================
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   policyholder_id  300 non-null    int64
 1   age              300 non-null    int64
```

```
 2   gender           300 non-null    str
 3   region           300 non-null    str
 4   vehicle_type     300 non-null    str
 5   annual_premium   300 non-null    float64
 6   claims_count     300 non-null    int64
 7   claim_amount     300 non-null    float64
dtypes: float64(2), int64(3), str(3)
memory usage: 18.9 KB

Statistical Summary:
       policyholder_id         age  annual_premium  claims_count  claim_amount
count       300.000000  300.000000      300.000000    300.000000    300.000000
mean        150.500000   45.876667     1842.409800      1.156667   1021.862000
std          86.746758   16.870988      869.436167      1.111865   1385.938919
min           1.000000   18.000000      418.540000      0.000000      0.000000
25%          75.750000   31.750000     1227.600000      0.000000      0.000000
50%         150.500000   45.000000     1691.510000      1.000000    651.515000
75%         225.250000   60.250000     2323.035000      2.000000   1316.285000
max         300.000000   75.000000     5013.770000      6.000000   9309.670000
```

[5]:

```python
# Explore column names and data types
print("Column Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

# Check for missing values
print("\nMissing Values:")
print(df.isna().sum())

# Examine unique values in categorical columns
print("\nUnique Regions:")
print(df['region'].unique())

print("\nUnique Vehicle Types:")
print(df['vehicle_type'].unique())

print("\nGender Distribution:")
print(df['gender'].value_counts())
```

```
Column Names:
['policyholder_id', 'age', 'gender', 'region', 'vehicle_type', 'annual_premium',
'claims_count', 'claim_amount']

Data Types:
policyholder_id      int64
age                  int64
```

```
gender                 str
region                 str
vehicle_type           str
annual_premium     float64
claims_count         int64
claim_amount       float64
dtype: object

Missing Values:
policyholder_id    0
age                0
gender             0
region             0
vehicle_type       0
annual_premium     0
claims_count       0
claim_amount       0
dtype: int64

Unique Regions:
<StringArray>
['East', 'West', 'Central', 'North', 'South']
Length: 5, dtype: str

Unique Vehicle Types:
<StringArray>
['Mid-range', 'SUV', 'Compact', 'Luxury']
Length: 4, dtype: str

Gender Distribution:
gender
Male          152
Female        135
Non-binary     13
Name: count, dtype: int64
```

1.3.6 2.4 Reading and Writing CSV Files

One of the most common tasks in data analysis is **reading data from files** and **writing results back**. Pandas makes this remarkably straightforward with `to_csv()` and `read_csv()`.

Method	Description	Example
`df.to_csv('file.csv', index=False)`	Write DataFrame to a CSV file	Export for sharing or archiving
`pd.read_csv('file.csv')`	Read a CSV file into a DataFrame	Load external data for analysis

Tip: Use `index=False` when writing to avoid saving the DataFrame index as an extra

column.

[6]:
```
# Write the insurance DataFrame to a CSV file
df.to_csv('insurance_sample.csv', index=False)
print('Data written to insurance_sample.csv')

# Read the CSV file back into a new DataFrame
df_loaded = pd.read_csv('insurance_sample.csv')

# Verify that the loaded data matches the original
print(f'Original shape:  {df.shape}')
print(f'Loaded shape:    {df_loaded.shape}')
print(f'Data identical:  {df.equals(df_loaded)}')
df_loaded.head(3)
```

```
Data written to insurance_sample.csv
Original shape:  (300, 8)
Loaded shape:    (300, 8)
Data identical:  True
```

[6]:
```
   policyholder_id  age  gender   region vehicle_type  annual_premium  \
0                1   56  Female     East    Mid-range         1755.21
1                2   69    Male     West    Mid-range         1344.78
2                3   46  Female  Central          SUV          590.09

   claims_count  claim_amount
0             1        811.11
1             0          0.00
2             2       2594.37
```

1.3.7 2.6 Exercise: Pandas Data Exploration

You will now explore the insurance dataset using Pandas methods. The data has already been loaded. Perform the following tasks:

1. **Examine the dataset structure** using `.head()`, `.info()`, and `.describe()`
2. **Analyze categorical columns** using `.unique()` and `.value_counts()`
3. **Compute grouped aggregations**:
   - Average premium by region
   - Average claims count by vehicle type
   - Count of policyholders by gender

Instructions: - Use the provided code cells below - Experiment with different grouping and aggregation combinations - Interpret the results in an actuarial context

[7]:
```
# Task 1: Get basic statistics for selected columns
print("Age Statistics:")
print(df['age'].describe())
```

```python
print("\nPremium Statistics:")
print(df['annual_premium'].describe())

# Task 2: Analyze the relationship between age groups and claims
# Create age bins
df['age_group'] = pd.cut(df['age'], bins=[0, 30, 40, 50, 100], labels=['<30',␣
↪'30-40', '40-50', '50+'])

print("\nPolicyholders by Age Group:")
print(df['age_group'].value_counts().sort_index())
```

```
Age Statistics:
count    300.000000
mean      45.876667
std       16.870988
min       18.000000
25%       31.750000
50%       45.000000
75%       60.250000
max       75.000000
Name: age, dtype: float64

Premium Statistics:
count     300.000000
mean     1842.409800
std       869.436167
min       418.540000
25%      1227.600000
50%      1691.510000
75%      2323.035000
max      5013.770000
Name: annual_premium, dtype: float64

Policyholders by Age Group:
age_group
<30       69
30-40     47
40-50     60
50+      124
Name: count, dtype: int64
```

[8]:
```python
# Task 3: Group and aggregate - Premium by Region
premium_by_region = df.groupby('region')['annual_premium'].agg(['mean',␣
↪'median', 'count'])
print("Premium Analysis by Region:")
print(premium_by_region)
```

```python
# Task 4: Group and aggregate - Claims count by Vehicle Type
claims_by_vehicle = df.groupby('vehicle_type')['claims_count'].agg(['mean',␣
↪'sum', 'count'])
print("\nClaims Analysis by Vehicle Type:")
print(claims_by_vehicle)

# Task 5: Group and aggregate - Multiple columns
region_gender_premium = df.groupby(['region', 'gender'])['annual_premium'].
↪mean().unstack()
print("\nAverage Premium by Region and Gender:")
print(region_gender_premium)
```

```
Premium Analysis by Region:
                mean    median  count
region
Central  1945.722833  1823.145     60
East     1758.096949  1555.610     59
North    1857.658308  1734.240     65
South    1833.275238  1654.780     63
West     1811.466415  1696.870     53

Claims Analysis by Vehicle Type:
                  mean  sum  count
vehicle_type
Compact       1.298851  113     87
Luxury        1.268293   52     41
Mid-range     0.950980   97    102
SUV           1.214286   85     70

Average Premium by Region and Gender:
gender        Female         Male   Non-binary
region
Central  1966.006552  1947.482222  1786.790000
East     1601.273929  1899.743548          NaN
North    1808.545667  1875.566061  2298.870000
South    1783.971000  1871.433125  2091.350000
West     1821.082222  1853.997586  1577.051667
```

1.4 Part 3: Plotting with Pandas

Pandas integrates seamlessly with Matplotlib, making it easy to create visualizations directly from DataFrames. Common plot types include:

1.4.1 3.1 Scatter Plots

Scatter plots show the relationship between two continuous variables. In insurance, we might visualize: - Age vs. Premium (does older age command higher premiums?) - Claims count vs. Claim amount (do frequent claimants have larger claims?)

```
df.plot.scatter(x='age', y='annual_premium', alpha=0.6)
```

The `alpha` parameter controls transparency, useful when points overlap.

1.4.2 3.2 Bar Charts

Bar charts compare categorical data. Common applications: - Average premium by region - Claim frequency by vehicle type - Policyholder counts by gender

```
df.groupby('region')['annual_premium'].mean().plot(kind='bar')
```

1.4.3 3.3 Histograms

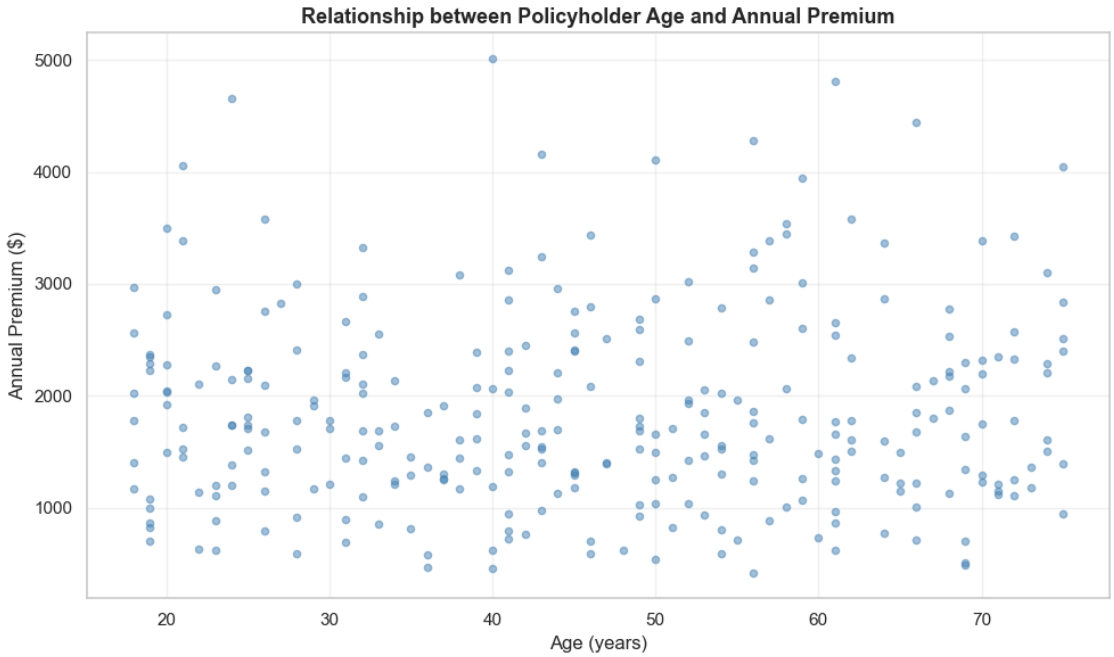
Histograms show the distribution of a continuous variable: - Premium distributions (are they normally distributed?) - Claim amounts (are they skewed?) - Age distribution of portfolio

```
df['annual_premium'].plot.hist(bins=30, edgecolor='black')
```

1.4.4 Key Visualization Principles

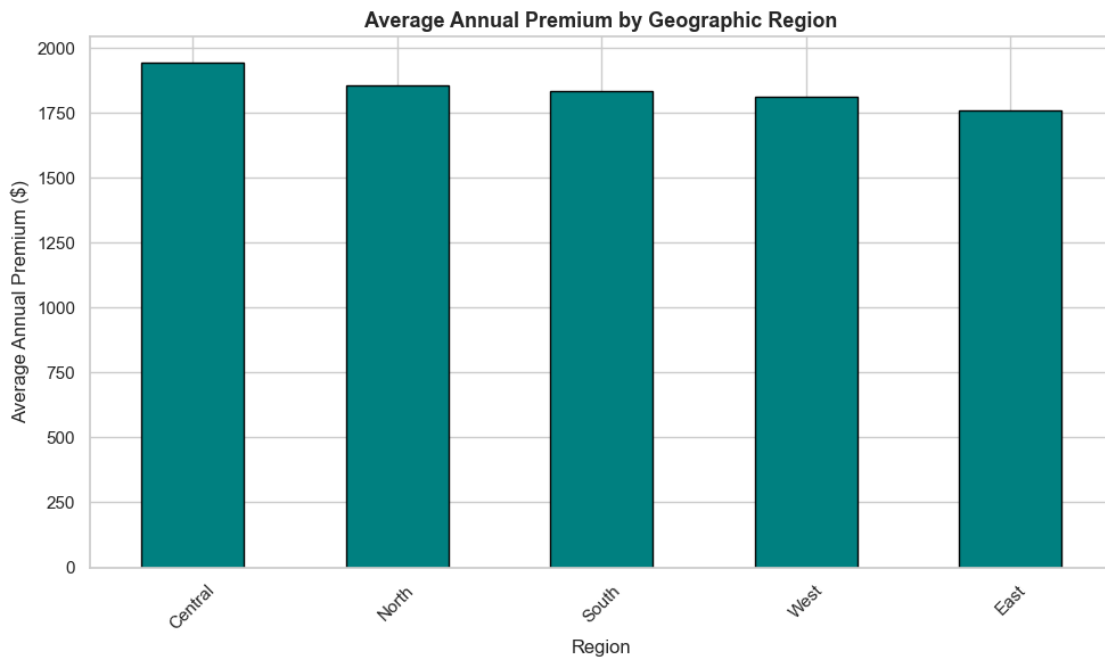
1. **Label axes clearly** with units
2. **Use informative titles** that state what is being shown
3. **Choose appropriate plot types** for your data
4. **Consider color and transparency** for clarity
5. **Add legends** when multiple series are shown

[9]:
```
# Plot 1: Scatter plot - Age vs. Premium
fig, ax = plt.subplots(figsize=(10, 6))
df.plot.scatter(x='age', y='annual_premium', alpha=0.5, ax=ax,␣
↪color='steelblue')
ax.set_xlabel('Age (years)', fontsize=12)
ax.set_ylabel('Annual Premium ($)', fontsize=12)
ax.set_title('Relationship between Policyholder Age and Annual Premium',␣
↪fontsize=13, fontweight='bold')
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print("Correlation between age and premium:", df['age'].
↪corr(df['annual_premium']).round(3))
```

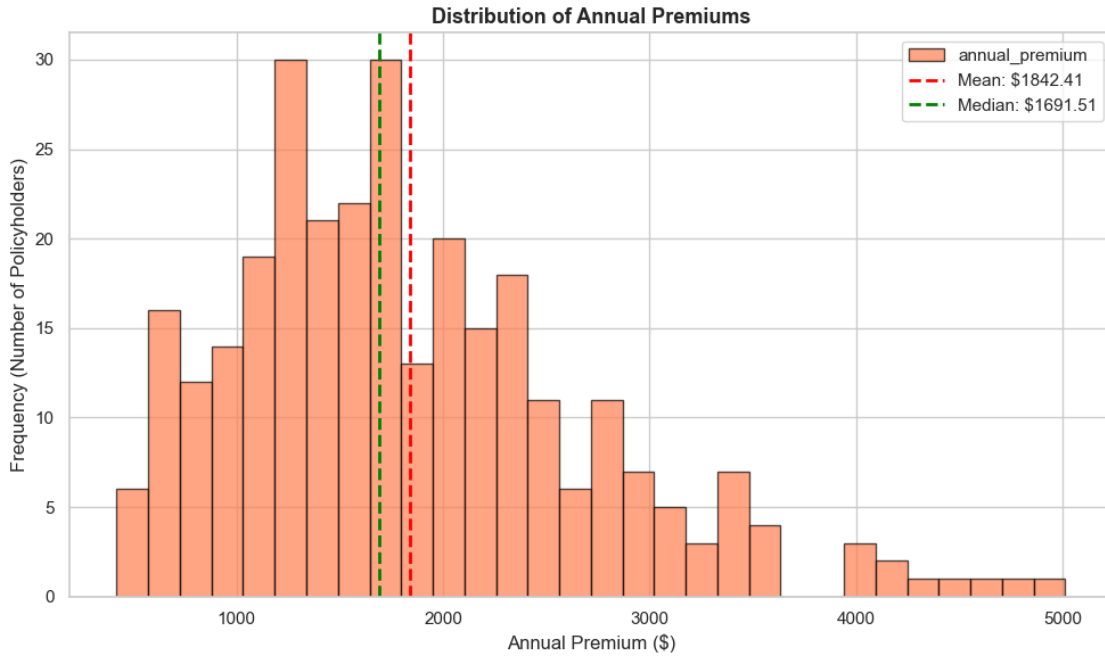



```
Correlation between age and premium: 0.033
```

[10]:
```
# Plot 2: Bar chart - Average Premium by Region
fig, ax = plt.subplots(figsize=(10, 6))
avg_premium_region = df.groupby('region')['annual_premium'].mean().
 ↪sort_values(ascending=False)
avg_premium_region.plot(kind='bar', ax=ax, color='teal', edgecolor='black')
ax.set_xlabel('Region', fontsize=12)
ax.set_ylabel('Average Annual Premium ($)', fontsize=12)
ax.set_title('Average Annual Premium by Geographic Region', fontsize=13,␣
 ↪fontweight='bold')
ax.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```



[11]:
```
# Plot 3: Histogram - Distribution of Premium Amounts
fig, ax = plt.subplots(figsize=(10, 6))
df['annual_premium'].plot.hist(bins=30, ax=ax, color='coral',␣
↪edgecolor='black', alpha=0.7)
ax.set_xlabel('Annual Premium ($)', fontsize=12)
ax.set_ylabel('Frequency (Number of Policyholders)', fontsize=12)
ax.set_title('Distribution of Annual Premiums', fontsize=13, fontweight='bold')
ax.axvline(df['annual_premium'].mean(), color='red', linestyle='--',␣
↪linewidth=2, label=f'Mean: ${df["annual_premium"].mean():.2f}')
ax.axvline(df['annual_premium'].median(), color='green', linestyle='--',␣
↪linewidth=2, label=f'Median: ${df["annual_premium"].median():.2f}')
ax.legend()
plt.tight_layout()
plt.show()

print(f"Mean Premium: ${df['annual_premium'].mean():.2f}")
print(f"Median Premium: ${df['annual_premium'].median():.2f}")
print(f"Std Dev: ${df['annual_premium'].std():.2f}")
```




```
Mean Premium: $1842.41
Median Premium: $1691.51
Std Dev: $869.44
```

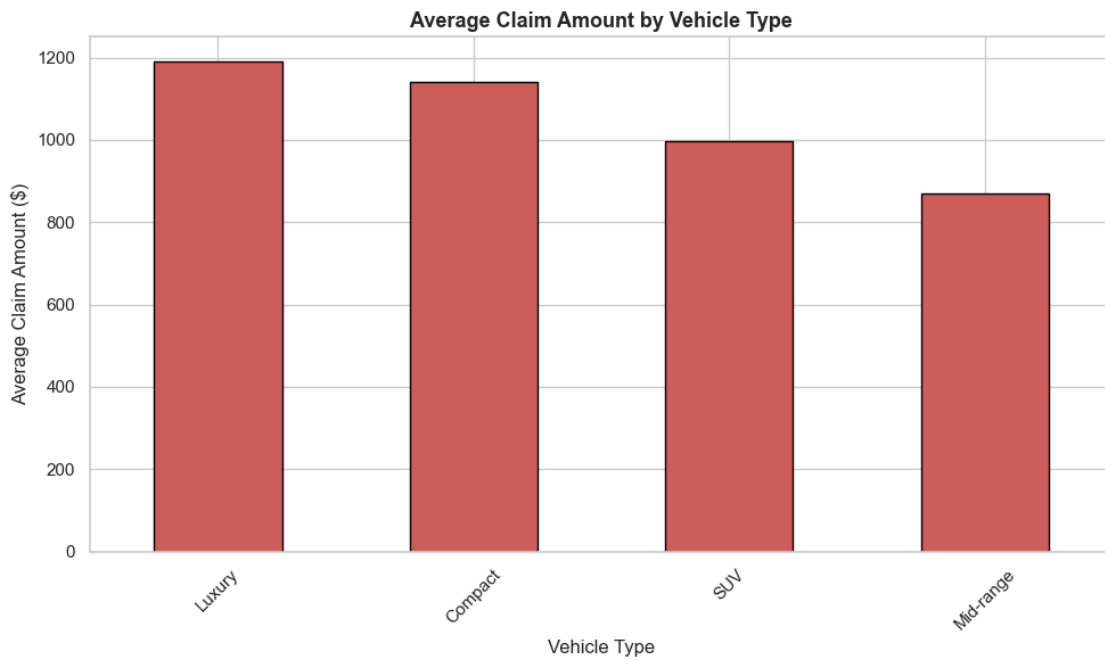
1.4.5 3.4 Exercise: Creating Pandas Visualizations

Create visualizations that answer these actuarial questions:

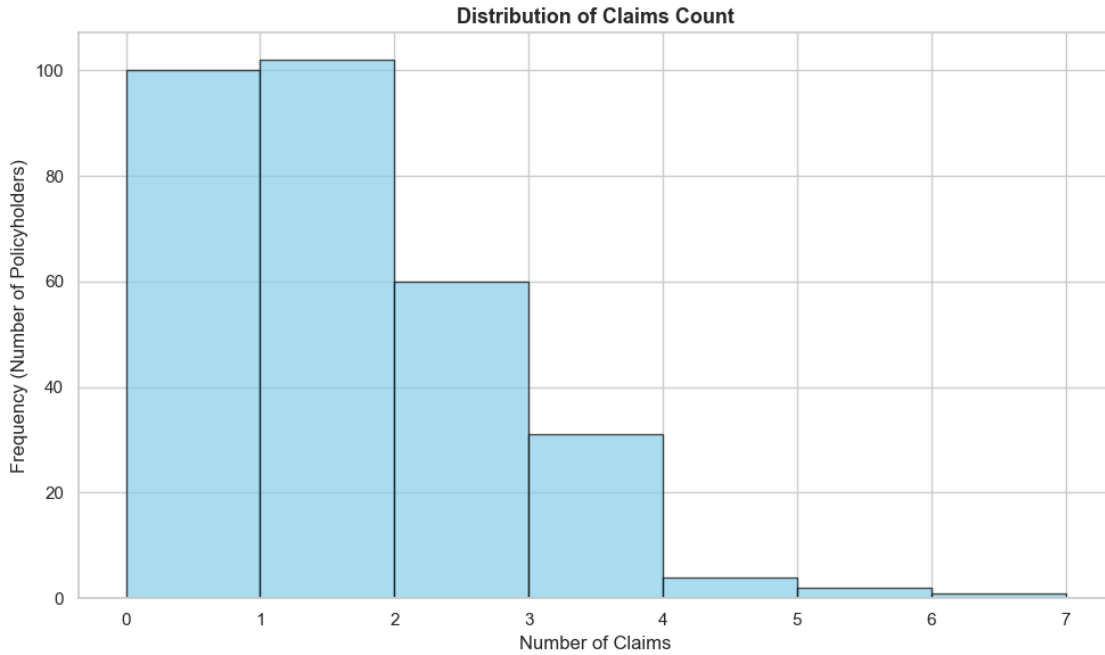
1. **How does claim amount vary by vehicle type?** (Bar chart)
2. **What is the distribution of claims count?** (Histogram)
3. **Is there a relationship between age and claims count?** (Scatter plot)

Instructions: - Write code to create each visualization - Add appropriate titles, labels, and legends - Interpret the results in an insurance context

[12]:
```
# Task 1: Bar chart - Average Claim Amount by Vehicle Type
fig, ax = plt.subplots(figsize=(10, 6))
avg_claim_vehicle = df.groupby('vehicle_type')['claim_amount'].mean().
 ↪sort_values(ascending=False)
avg_claim_vehicle.plot(kind='bar', ax=ax, color='indianred', edgecolor='black')
ax.set_xlabel('Vehicle Type', fontsize=12)
ax.set_ylabel('Average Claim Amount ($)', fontsize=12)
ax.set_title('Average Claim Amount by Vehicle Type', fontsize=13,␣
 ↪fontweight='bold')
ax.tick_params(axis='x', rotation=45)
plt.tight_layout()
plt.show()
```

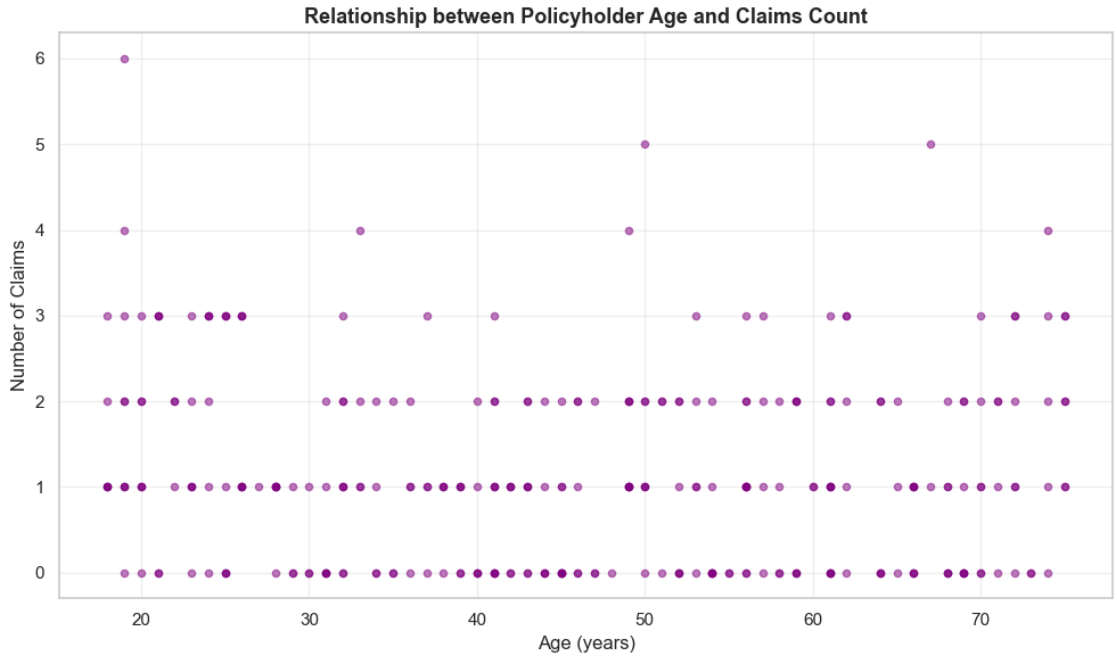


[13]:
```
# Task 2: Histogram - Distribution of Claims Count
fig, ax = plt.subplots(figsize=(10, 6))
df['claims_count'].plot.hist(bins=range(0, int(df['claims_count'].max())+2),␣
↪ax=ax, color='skyblue', edgecolor='black', alpha=0.7)
ax.set_xlabel('Number of Claims', fontsize=12)
ax.set_ylabel('Frequency (Number of Policyholders)', fontsize=12)
ax.set_title('Distribution of Claims Count', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()

print(f"Average claims per policyholder: {df['claims_count'].mean():.2f}")
print(f"Max claims: {df['claims_count'].max()}")
```




```
Average claims per policyholder: 1.16
Max claims: 6
```

[14]:
```
# Task 3: Scatter plot - Age vs. Claims Count
fig, ax = plt.subplots(figsize=(10, 6))
df.plot.scatter(x='age', y='claims_count', alpha=0.5, ax=ax, color='purple')
ax.set_xlabel('Age (years)', fontsize=12)
ax.set_ylabel('Number of Claims', fontsize=12)
ax.set_title('Relationship between Policyholder Age and Claims Count',␣
↪fontsize=13, fontweight='bold')
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

print("Correlation between age and claims count:", df['age'].
↪corr(df['claims_count']).round(3))
```




```
Correlation between age and claims count: -0.076
```

1.5 Part 4: Advanced Visualization with Seaborn

Seaborn is a statistical visualization library built on Matplotlib. It provides elegant, high-level interfaces for creating complex visualizations with minimal code.

1.5.1 4.1 Distribution Visualization

Box Plots Box plots summarize the distribution of a variable by showing: - **Median** (center line) - **Interquartile Range** (box: 25th to 75th percentile) - **Whiskers** (typically extend to 1.5 × IQR) - **Outliers** (individual points)

Box plots are ideal for comparing distributions across groups, such as premiums by vehicle type.

Violin Plots Violin plots combine a box plot with a kernel density estimate (KDE). They show: - The full distribution shape - Density at different values - More information than box plots but can be harder to interpret at first glance

1.5.2 4.2 Pairplots

A pairplot shows relationships between multiple variables simultaneously. It creates: - Scatter plots for every pair of numeric variables - Histograms on the diagonal

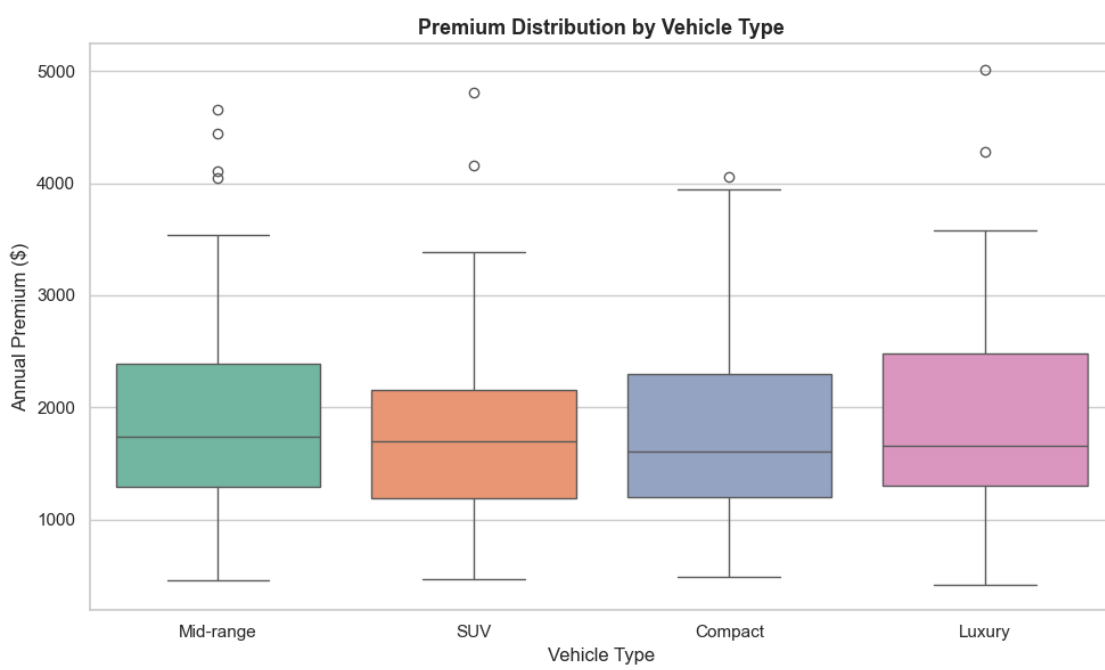
This is powerful for exploratory data analysis and identifying patterns in multivariate data.

1.5.3 Actuarial Application

In insurance analytics, these visualizations help: - **Identify outliers** (unusual claims or premiums) - **Compare groups** (e.g., do different regions have different risk profiles?) - **Find relationships** (do multiple variables move together?) - **Communicate findings** to non-technical stakeholders

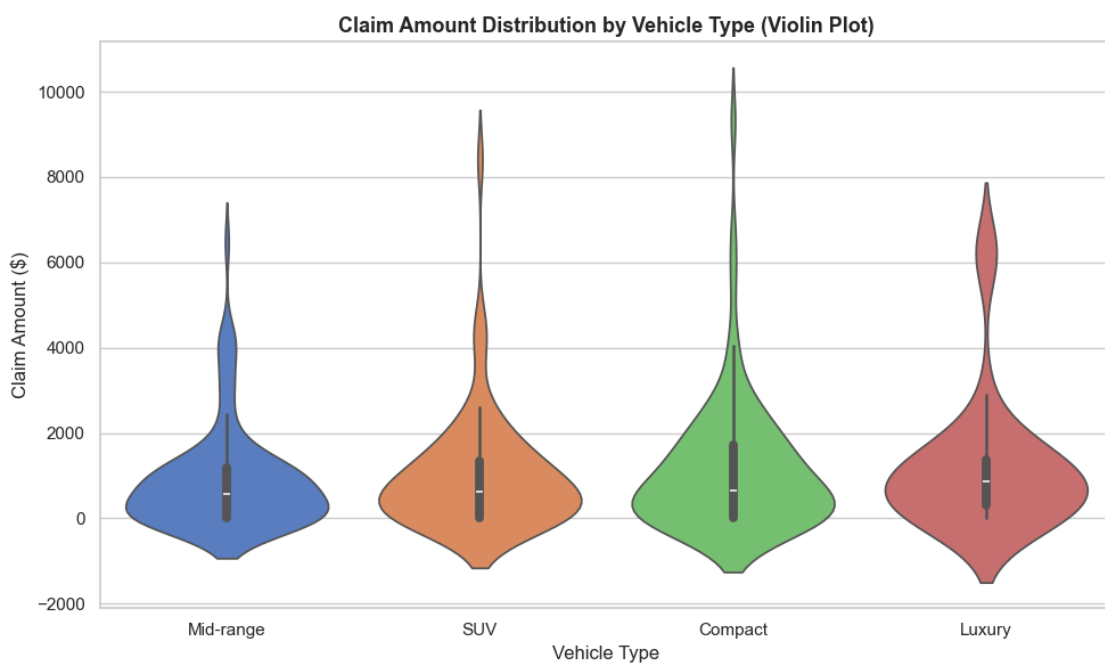
[15]:
```
# Plot 1: Box plot - Premium distribution by Vehicle Type
fig, ax = plt.subplots(figsize=(10, 6))
sns.boxplot(data=df, x='vehicle_type', y='annual_premium', palette='Set2',␣
↪ax=ax)
ax.set_xlabel('Vehicle Type', fontsize=12)
ax.set_ylabel('Annual Premium ($)', fontsize=12)
ax.set_title('Premium Distribution by Vehicle Type', fontsize=13,␣
↪fontweight='bold')
plt.tight_layout()
plt.show()
```

```
/var/folders/pk/cwgxjc893d9149wsd8y22c9r0000gn/T/ipykernel_85020/2280384260.py:3
: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(data=df, x='vehicle_type', y='annual_premium', palette='Set2',
ax=ax)
```



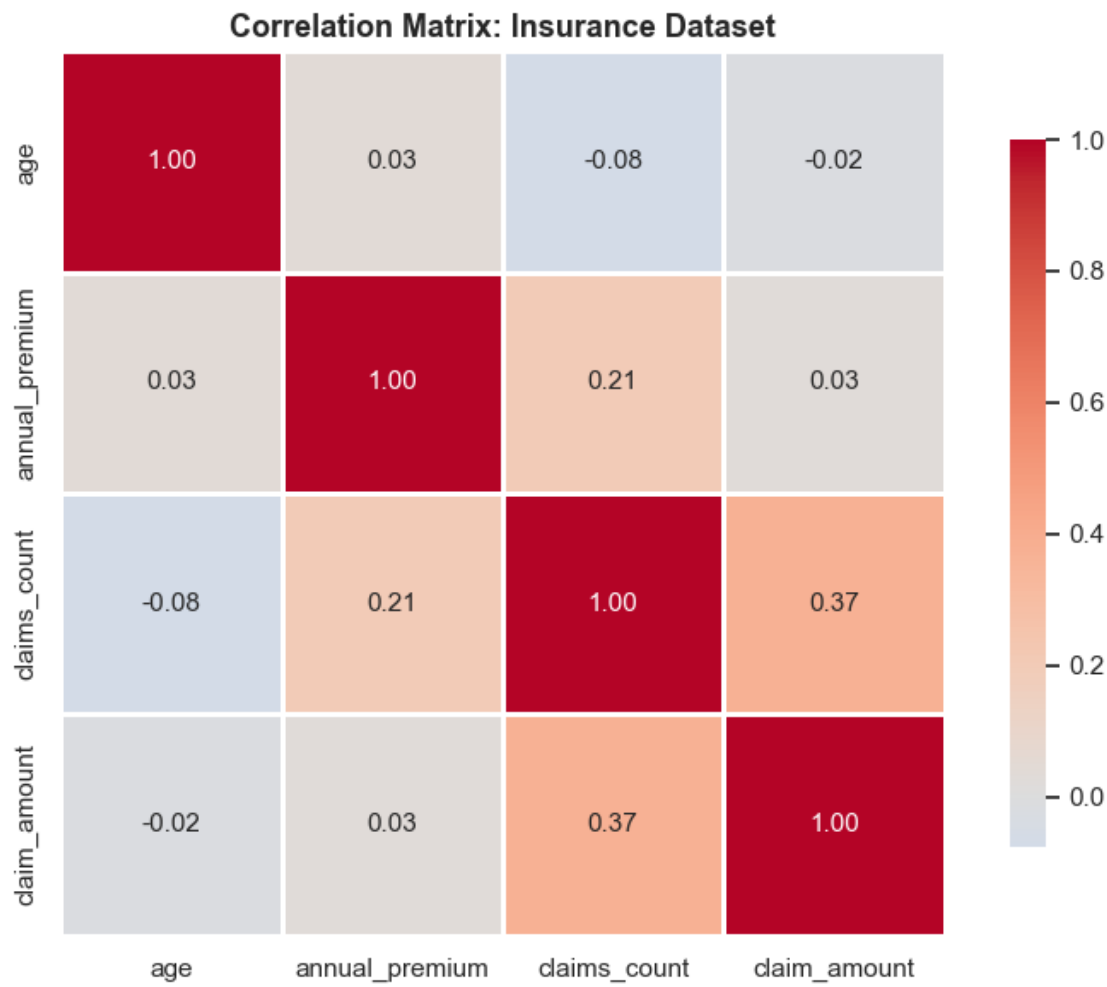
[16]:
```
# Plot 2: Violin plot - Claim Amount distribution by Vehicle Type
fig, ax = plt.subplots(figsize=(10, 6))
sns.violinplot(data=df, x='vehicle_type', y='claim_amount', palette='muted',␣
↪ax=ax)
ax.set_xlabel('Vehicle Type', fontsize=12)
ax.set_ylabel('Claim Amount ($)', fontsize=12)
ax.set_title('Claim Amount Distribution by Vehicle Type (Violin Plot)',␣
↪fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
```

```
/var/folders/pk/cwgxjc893d9149wsd8y22c9r0000gn/T/ipykernel_85020/3346576513.py:3
: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=df, x='vehicle_type', y='claim_amount', palette='muted',
ax=ax)
```



[17]:
```
# Plot 3: Heatmap - Correlation matrix
# Select numeric columns
numeric_df = df[['age', 'annual_premium', 'claims_count', 'claim_amount']]
corr_matrix = numeric_df.corr()
```

```
fig, ax = plt.subplots(figsize=(8, 6))
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm', center=0,␣
↪ax=ax,
            square=True, linewidths=1, cbar_kws={"shrink": 0.8})
ax.set_title('Correlation Matrix: Insurance Dataset', fontsize=13,␣
↪fontweight='bold')
plt.tight_layout()
plt.show()

print("Correlation Matrix:")
print(corr_matrix)
```




```
Correlation Matrix:
                     age  annual_premium  claims_count  claim_amount
age             1.000000        0.033249     -0.076346     -0.021455
annual_premium  0.033249        1.000000      0.205191      0.029816
```

```
claims_count    -0.076346        0.205191       1.000000       0.370306
claim_amount    -0.021455        0.029816       0.370306       1.000000
```

[18]:
```
# Plot 4: Pairplot - Relationships between all numeric variables
# Create a subset with numeric columns and one categorical variable for color
pairplot_df = df[['age', 'annual_premium', 'claims_count', 'claim_amount',␣
↪'vehicle_type']].copy()

# For better visualization, create pairplot with a subset (to keep it␣
↪manageable)
g = sns.pairplot(pairplot_df, hue='vehicle_type', palette='husl',␣
↪diag_kind='hist',
                 plot_kws={'alpha': 0.6, 's': 50}, height=2.5)
g.fig.suptitle('Pairplot: Relationships in Insurance Data', fontsize=14,␣
↪fontweight='bold', y=1.00)
plt.tight_layout()
plt.show()
```

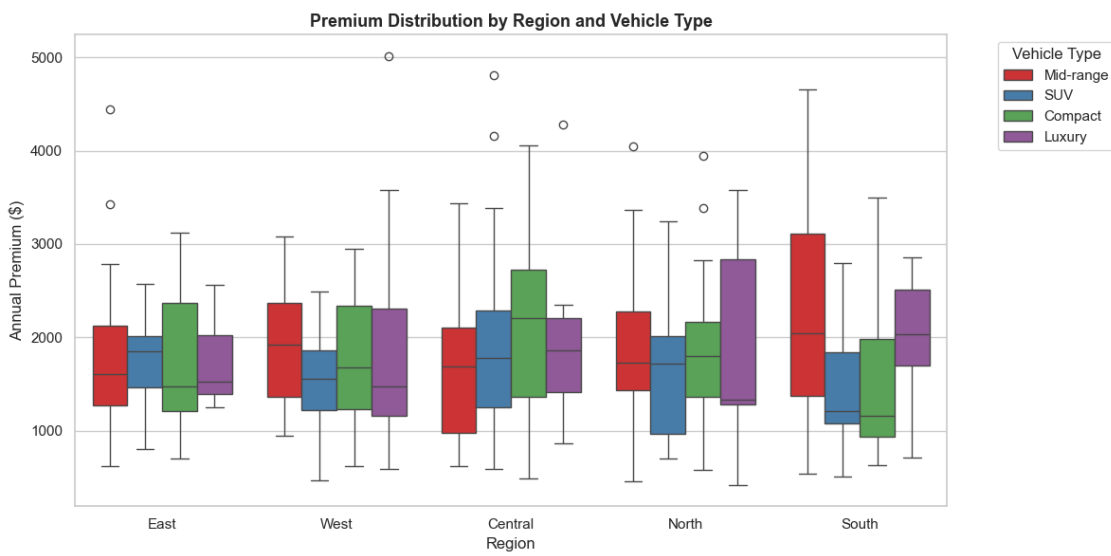
1.5.4 4.3 Exercise: Seaborn Visualization Mastery

Create advanced visualizations to explore relationships in the insurance data:

1. **Box plot comparison**: Compare premiums across regions and vehicle types
2. **Distribution analysis**: Create a violin plot for claim amounts by gender
3. **Correlation exploration**: Identify which variables move together most strongly

Instructions: - Use Seaborn functions for elegant, informative visualizations - Add clear titles and labels - Discuss what patterns you observe and their actuarial significance

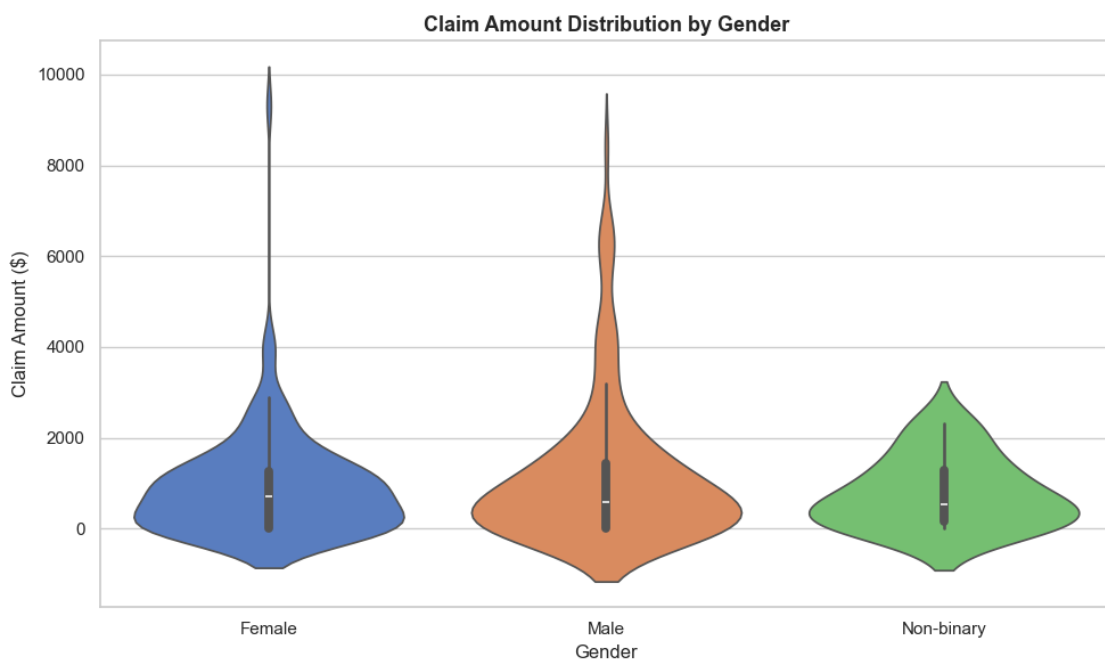
[19]:
```
# Task 1: Box plot - Premium by Region and Vehicle Type
fig, ax = plt.subplots(figsize=(12, 6))
sns.boxplot(data=df, x='region', y='annual_premium', hue='vehicle_type',␣
↪palette='Set1', ax=ax)
ax.set_xlabel('Region', fontsize=12)
ax.set_ylabel('Annual Premium ($)', fontsize=12)
ax.set_title('Premium Distribution by Region and Vehicle Type', fontsize=13,␣
↪fontweight='bold')
ax.legend(title='Vehicle Type', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



[20]:
```
# Task 2: Violin plot - Claim Amount by Gender
fig, ax = plt.subplots(figsize=(10, 6))
sns.violinplot(data=df, x='gender', y='claim_amount', palette='muted', ax=ax)
ax.set_xlabel('Gender', fontsize=12)
ax.set_ylabel('Claim Amount ($)', fontsize=12)
```

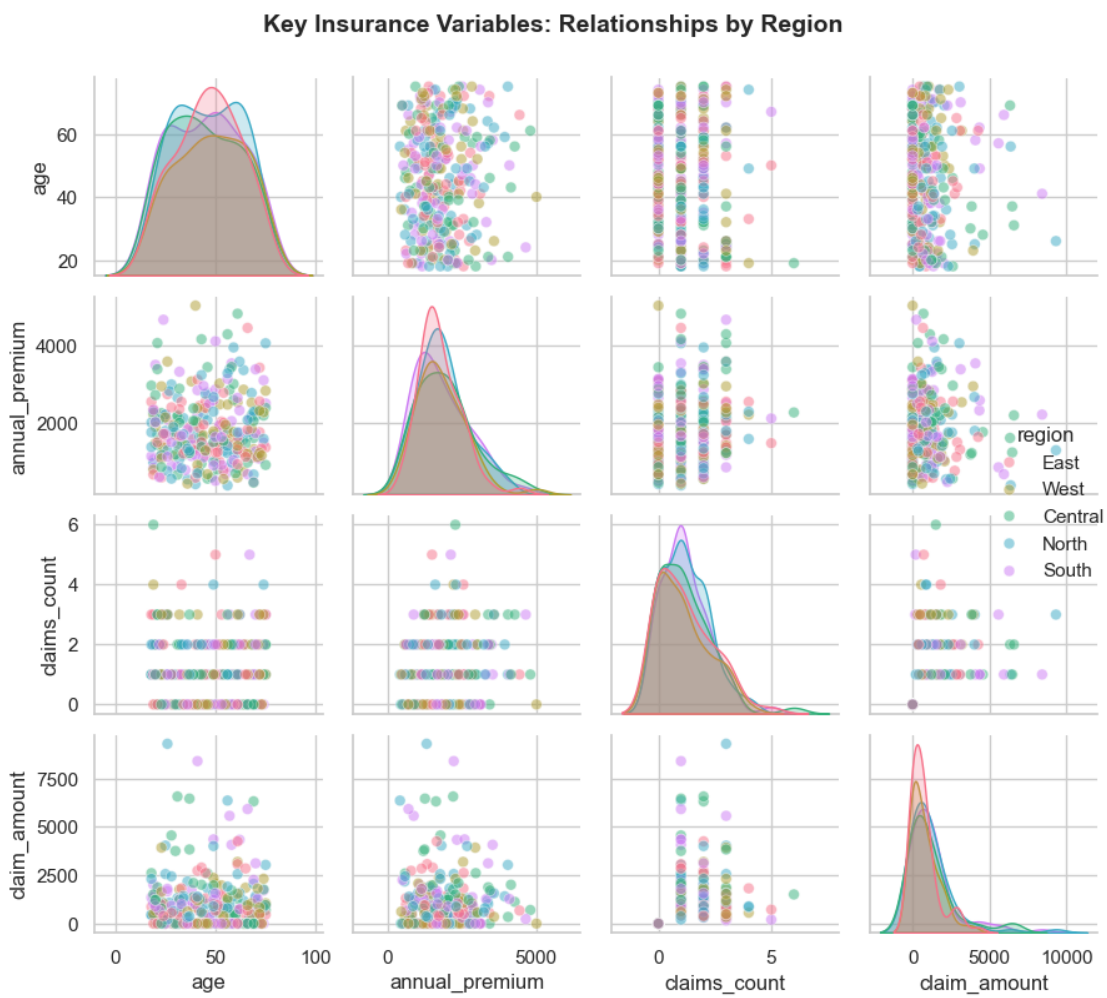
```
ax.set_title('Claim Amount Distribution by Gender', fontsize=13,␣
 ↪fontweight='bold')
plt.tight_layout()
plt.show()

# Summary statistics by gender
print("Average claim amount by gender:")
print(df.groupby('gender')['claim_amount'].agg(['mean', 'median', 'std',␣
 ↪'count']))
```

```
/var/folders/pk/cwgxjc893d9149wsd8y22c9r0000gn/T/ipykernel_85020/4117814105.py:3
: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=df, x='gender', y='claim_amount', palette='muted', ax=ax)
```




```
Average claim amount by gender:
                   mean  median          std  count
gender
Female       906.053037  694.98  1153.848827    135
Male        1145.824145  582.11  1594.549594    152
Non-binary   775.090000  529.80   766.766807     13
```

[21]:
```
# Task 3: Create a focused pairplot with subset of key variables
# Select key variables for analysis
key_vars = df[['age', 'annual_premium', 'claims_count', 'claim_amount',␣
↪'region']].copy()

# Create pairplot with region coloring
g = sns.pairplot(key_vars, hue='region', palette='husl', diag_kind='kde',
                 plot_kws={'alpha': 0.5, 's': 40}, height=2)
g.fig.suptitle('Key Insurance Variables: Relationships by Region', fontsize=14,␣
↪fontweight='bold', y=1.00)
plt.tight_layout()
plt.show()
```



1.6 Summary and Key Takeaways

Congratulations! You have now completed a comprehensive introduction to data visualization for insurance analytics using Python.

1.6.1 What You Have Learned

1. **NumPy Fundamentals**
   - Create and manipulate arrays efficiently
   - Apply vectorized operations to large datasets
   - Work with multi-dimensional data
2. **Pandas Data Manipulation**
   - Load and explore data using DataFrames and Series
   - Perform grouping and aggregation operations
   - Analyze insurance data systematically
3. **Visualization with Pandas**
   - Create scatter plots to explore relationships
   - Use bar charts for categorical comparisons
   - Build histograms to understand distributions
4. **Advanced Visualization with Seaborn**
   - Leverage box plots and violin plots for distribution analysis
   - Build pairplots to investigate multivariate relationships
   - Create heatmaps to visualize correlations

1.6.2 Actuarial Applications

The techniques covered in this session have direct applications to actuarial work:

- **Premium Modeling**: Understand how policyholder characteristics affect premiums
- **Claims Analysis**: Identify patterns in claim frequency and severity
- **Risk Segmentation**: Compare risk profiles across different groups
- **Portfolio Analysis**: Visualize portfolio composition and characteristics
- **Data Quality**: Detect outliers and data issues early

1.6.3 Next Steps

To deepen your skills: 1. Practice with your own datasets 2. Explore advanced statistical methods (regression, clustering) 3. Learn about hypothesis testing and confidence intervals 4. Study machine learning approaches for prediction

1.6.4 Resources

- NumPy Documentation
- Pandas Documentation
- Matplotlib User Guide
- Seaborn Tutorial

Thank you for participating in this workshop! We hope you now feel confident using Python for data visualization and analysis in your actuarial work.